The ϕ^4 field as a second laboratory

The derivation, the estimator, the measurements, and the confirmatory results

Jérôme's email of 1 October 2026 (after Marc) · this report 1 October 2026 · branch `claude/phi4-field-2026-10-01` · revised 1 October (reversibility, the physicist's parametrisation) and 3–5 October (every confirmatory result, E35, E33's amendment 3)

Summary

Jérôme and Marc asked for a test of the estimator on Markov data from a physical model: spatial snapshots of the one-dimensional stochastic Allen–Cahn (ϕ^4) field, whose stationary law is a nearest-neighbour Gibbs measure, with four arms tracked across temperature and lattice spacing (a learned pair energy, the Gaussian closure, the window CNN, the oracle score), and the suggestion that with this concrete example the paper may be complete. Everything in this report was built, tested and registered on 1 October; by 5 October every registered run had landed, been read against its predictions and been frozen (§6).

1. **The chain is exact and in the package** (§1). Its kernel is a Gaussian step tilted by a function of the *arrival* point, the column normaliser being the ground-state transform. The Gaussian linear chain is the one member the two laboratories share (Proposition 3); the paper's mixture kernel is misspecified here through a nonlinear drift, not through the shape of its driving noise.
2. **The learned pair energy is a transition kernel** (§2): concave M-step, the gradient is Boltzmann learning's positive phase minus a *conditional* negative phase, no chain-wide partition function (Proposition 6).
3. **The directed kernel and the symmetric pair energy are one model class** (Proposition 7, proved this afternoon): every tilted walk is reversible, with invariant law gZ and site energy $f = -\frac{1}{2}\log(g/Z)$, which for the field is the potential. A fitted kernel therefore reads as physics, $\hat{\Theta} = \hat{\gamma}s^2/h$ and $\hat{V}$ from $\hat{f}$ (Corollary 8); and the parsimonious exactly specified model parametrises f itself: the quartic site energy, five parameters, Jérôme's W_θ literally (§2.1). It joins E33 as a fifth arm under a registered amendment with two further predictions.
4. **One thing in the email does not survive** (§3): moment projection of the messages is not a well-posed estimator on this chain, because the backward message is nearly flat. Measured: 42–100 % of a projected message's mass at the grid edge. The Gaussian-closure arm is therefore the covariance twin, which is what Proposition 1 of the paper proves projection computes on linear chains.
5. **E32** (§4): at matched correlation $\lambda_1/\lambda_0 = 0.85$ the linear floor is 0.269 at $\Theta = 0.3$ against 0.168 for the Laplace laboratory, and falls to 0.041 at $\Theta = 2$; the operating grid's score agrees with a four times finer grid's to 10^{-12}; Theorem 2's gap ratio rises monotonically through t_* to 0.73–0.94 at $t = 3$.
6. **E33 and E34** (§5): five kernel arms through grid BP, the window head and the transformer under both interfaces, the oracles; the physical statistics the consolidated note asks for; the clean-data interface. Eleven predictions were registered before any confirmatory cell ran; one Legendre fit crashed the kernel jobs on the finest lattice, and amendment 3 (a failed fit is a row counted against the prediction, never a crash) was registered before any kernel row was read.
7. **The confirmatory results** (§6). The five-parameter site energy is the best estimator on every cell at every size: through the noise it reaches 0.03–0.17 of the linear floor at $M = 2048$ and beats the 25-parameter spline by 43–59 %; it reads the temperature within 10 % in 96/96 fits (spread 0.3–2.3 %) and the barrier within 20 % in 78/80; its samples reproduce the kink density and domain length within 5.3 % and 2.9 %. The paper's mixture kernel converges onto the linear floor (within 2 % at $M = 2048$ on every cell), and its samples have up to 1.7 times too many kinks. The window head and the transformer are 4 to 27 times the spline's risk on every cell and, at $\Theta \geq 0.5$, worse than the linear floor. The Legendre tilt is unreliable: it parks its invariant law outside the data, is sometimes reducible on the grid, and failed 39 fits on the finest lattice. Under a global sign latent (E35) the first-order kernel is worse than the linear floor and the same kernel on the augmented state is 8.6 and 17.9 times better. Of E33's eight predictions, P1, P3, P4 and P6 are confirmed, P8 holds for the temperature and misses the barrier on one cell, and P2, P5 and P7 are falsified as written, P7 because the site energy did better than predicted.

Notation (one table for the code, the paper and this report: `docs/NOTATION.md`). Θ is the physical temperature (the code and the cell names call it `T`), h the lattice spacing, ϕ_i the field and $a_i = \phi_i/s$ its standardised value, t the diffusion time of the channel $x_t = e^{-t}a + \sqrt{\Delta_t}\,z$ with $\Delta_t = 1 - e^{-2t}$, L the number of sites, M the number of training sequences, N_g the grid size. The consolidated note's σ is our $\sigma_t = \sqrt{e^{2t} - 1}$ (its channel $Y = X + \sigma Z$ is ours multiplied by e^t). $K(a' \mid a)$ is always a normalised transition and $\mathsf{T} = e^{-W}$ the transfer operator; ℓ_i is the local likelihood, $\overrightarrow{m}_i$ and $\overleftarrow{m}_i$ the forward and backward messages without it (the note's $\alpha_i = \overrightarrow{m}_i\ell_i$ and $\beta_i = \overleftarrow{m}_i$); γ is the bond variance and ξ the correlation length. A linear chain $a_{i+1} = \rho a_i + \varepsilon_i$ has driving noise ε_i (the paper

used to say "innovation"). *Grid BP* is sum-product on $N_g = 401$ trapezoidal points on $[-8, 8]$: exact for the finite-state model the grid defines, an approximation of the continuous chain whose truncation and quadrature errors are measured (E32 P1); run with the true kernel it gives the *reference* (oracle) score.

1 The chain

1.1 From the SPDE to a Markov chain

The equation $\partial_\tau \phi = \partial_x^2 \phi - V'(\phi) + \sqrt{2\Theta}\,\eta$ with $V(\phi) = (\phi^2 - 1)^2/4$ is the gradient flow of $H[\phi] = \int [\frac{1}{2}(\partial_x \phi)^2 + V(\phi)]\,\mathrm{d}x$ at temperature Θ; in one space dimension its stationary law is the Gibbs measure $\propto \exp(-H/\Theta)$. On a lattice of spacing h, $H_h(\phi) = \sum_i [(\phi_{i+1} - \phi_i)^2/2h + hV(\phi_i)]$, a nearest-neighbour energy. A spatial snapshot is therefore a first-order Markov chain in the site index. Writing the site terms symmetrically,

$$\mathsf{T}(\phi, \phi') = D(\phi)\, G(\phi' - \phi)\, D(\phi'), \qquad D = e^{-hV/2\Theta}, \quad G(u) = e^{-u^2/2\Theta h}, \tag{1}$$

is a symmetric, positive kernel, Hilbert–Schmidt for a confining V, with a simple largest eigenvalue λ_0 and a positive eigenfunction ψ_0, $\int \psi_0^2 = 1$.

Proposition 1 (ground-state transform)**.** *$K(\phi' \mid \phi) = \mathsf{T}(\phi, \phi')\psi_0(\phi')/(\lambda_0 \psi_0(\phi))$ is a Markov kernel; $\pi = \psi_0^2$ is invariant for it; and the chain is reversible, $\pi(\phi)K(\phi' \mid \phi) = \pi(\phi')K(\phi \mid \phi')$.*

Proof. $\int K(\phi' \mid \phi)\,\mathrm{d}\phi' = \int \mathsf{T}(\phi, \phi')\psi_0(\phi')\,\mathrm{d}\phi'/(\lambda_0\psi_0(\phi)) = 1$ by the eigen-equation. $\int \pi(\phi)K(\phi' \mid \phi)\,\mathrm{d}\phi = \psi_0(\phi') \int \mathsf{T}(\phi, \phi')\psi_0(\phi)\,\mathrm{d}\phi/\lambda_0 = \psi_0(\phi')^2$ by symmetry of T and the eigen-equation again. $\pi(\phi)K(\phi' \mid \phi) = \psi_0(\phi)\mathsf{T}(\phi, \phi')\psi_0(\phi')/\lambda_0$ is symmetric in (ϕ, ϕ'). □

The infinite-volume Gibbs measure restricted to L consecutive sites is this stationary chain, $\pi(\phi_1) \prod K(\phi_{i+1} \mid \phi_i)$; that is what "spatial snapshots of a stationary field" means, and it is the law the data are drawn from. (A finite open chain with free ends is also Markov but inhomogeneous within a correlation length of its ends; we do not use it.) The lag-k covariance is $\sum_{n \geq 1} (\lambda_n/\lambda_0)^k \langle \psi_0 | \phi | \psi_n \rangle^2$, so $\rho_1 := \lambda_1/\lambda_0$ plays the role ρ plays for the linear chains and the correlation length in sites is $\xi = 1/\log(1/\rho_1)$. In one dimension there is no phase transition at any $\Theta > 0$: a kink costs the finite energy E_{kink}, so $\rho_1 < 1$ and ξ is finite at every temperature, growing roughly like $e^{E_{\mathrm{kink}}/\Theta}$ as $\Theta \to 0$. The low-temperature cells sit in a sharp crossover, not in an ordered phase.

1.2 The tilted-walk identity

Proposition 2 (tilted walk)**.** *With $g := D\psi_0$,*

$$K(\phi' \mid \phi) = \frac{G(\phi' - \phi)\, g(\phi')}{Z(\phi)}, \qquad Z(\phi) = \int G(\phi' - \phi) g(\phi')\,\mathrm{d}\phi' = \frac{\lambda_0 \psi_0(\phi)}{D(\phi)}. \tag{2}$$

Proof. From (1), $\mathsf{T}(\phi, \phi')\psi_0(\phi') = D(\phi)G(\phi' - \phi)g(\phi')$, and the eigen-equation reads $D(\phi) \int G(\phi' - \phi)g(\phi')\,\mathrm{d}\phi' = \lambda_0\psi_0(\phi)$, which is the stated Z. Substituting in Proposition 1's K gives (2). □

So the kernel is a Gaussian random-walk step of variance Θh, tilted by a positive function of the arrival point and normalised per column: Jérôme's "Gaussian increment tilted by $\psi_0(\phi')e^{-hV(\phi')/\Theta}/\psi_0(\phi)$", with the normaliser made explicit. The point for the estimator is that *the column normaliser is the ground-state transform*: a family that parametrises $\log g$ and normalises columns on the grid contains the truth exactly, and never needs ψ_0 or λ_0.

1.3 Where the two laboratories meet

Proposition 3 (Gaussian embedding)**.** *The linear-Gaussian kernel $\mathcal{N}(a'; \rho a, q)$ with $0 < \rho < 1$ is the member of (2) with step variance $\gamma = q/\rho$ and $\log g(a') = -(1 - \rho)a'^2/2q + \mathrm{const}$. No other kernel of the linear form $p_\varepsilon(a' - \rho a)$, $\rho \neq 0$, belongs to the family, and no other member of the family has state-independent driving noise.*

Proof. $\log \mathcal{N}(a'; \rho a, q) = -a'^2/2q + \rho a a'/q - \rho^2 a^2/2q + \mathrm{c}$ and $\log[G_\gamma(a' - a)g(a')] = -a'^2/2\gamma + aa'/\gamma - a^2/2\gamma + \log g(a') + \mathrm{c}$. Matching the cross term gives $\gamma = q/\rho$; the a'^2 terms give $\log g(a') = -a'^2(1/q - \rho/q)/2$, a Gaussian of variance $q/(1 - \rho) = 1 + \rho$; the a^2 terms are absorbed by $Z(a)$. For the second claim, $\partial_a \partial_{a'} \log K$ equals $1/\gamma$, a constant, for every member of (2), and equals $-\rho(\log p_\varepsilon)''(a' - \rho a)$ for a linear kernel, which is constant iff $\log p_\varepsilon$ is quadratic. □

Remark 4 (the misspecification mechanism)*.* Expanding $\log g$ around a, $K(a' \mid a) \approx \mathcal{N}\big(a';\, a + \gamma(\log g)'(a),\, \gamma\big)$: at fine h the chain is the Euler step of $\mathrm{d}a = \gamma(\log g)'(a)\,\mathrm{d}\tau + \sqrt{2\gamma}\,\mathrm{d}W_\tau$ with a *nonlinear* drift. The mixture kernel $\sum_c \pi_c \mathcal{N}(a' - \rho a; \mu_c, s_c^2)$ has a linear drift and state-independent driving noise; it is misspecified here through the drift. That is a different mechanism from every axis of the paper's misspecification study (global latent, long-range coupling, a second lag, site-dependent ρ, heteroskedasticity), and E33's P1/P2 measure it.

1.4 Standardisation, cells

Everything downstream assumes unit marginal variance, so the package exposes $a = \phi/s$, $s^2 = \mathrm{Var}_\pi(\phi)$ (the mean vanishes by symmetry); $K_a(a' \mid a) = sK_\phi(sa' \mid sa)$. The (Θ, h) map (Fig. 1) gives the iso-correlation line $h^*(\Theta)$ at $\rho_1 = 0.85$, the laboratory's frozen correlation: it passes through $\Theta = 0.30$ at $h = 1$. The temperature sweep sits on that line, so the second laboratory is compared with the first at matched second-order structure while the marginal kurtosis runs from -1.35 to -0.46; the refinement sweep halves h twice from the base cell with $L \propto 1/h$, so the physical length is fixed and the correlation length in sites grows $6 \to 12 \to 24$ (Table 1). The step width $\sqrt{\Theta h}/s$ is at least 14 grid cells on every cell, so the usual resolution condition is met with room.

cell	Θ	h	L	ρ_1	ξ (sites)	s	$\kappa_4(\pi)$
base	0.3	1	32	0.848	6.08	0.821	−1.352
T-sweep	0.5	0.419	32	0.850	6.15	0.766	−0.881
T-sweep	1.0	0.226	32	0.850	6.15	0.818	−0.588
T-sweep	2.0	0.150	32	0.850	6.15	0.950	−0.461
refine	0.3	0.5	64	0.917	11.47	0.805	−1.253
refine	0.3	0.25	128	0.957	22.48	0.800	−1.218

Table 1: The six confirmatory cells (`outputs/E32/cells.json`). h^* solved by bisection to $\rho_1 = 0.85 \pm 10^{-4}$; $\xi = 1/\log(1/\rho_1)$.

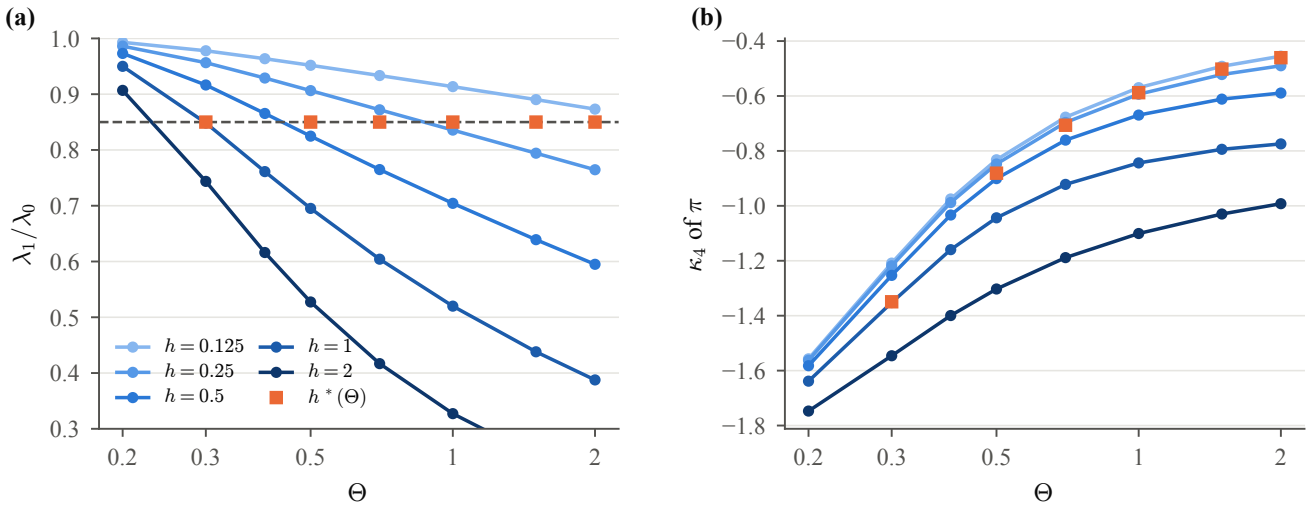


Figure 1: The (Θ, h) map of the chain. (a) λ_1/λ_0 against Θ for five spacings; the dashed line is the laboratory's $\rho = 0.85$ and the squares are the solved $h^*(\Theta)$. (b) Excess kurtosis of the standardised marginal: the non-Gaussianity knob, platykurtic throughout (bimodal), from -1.75 to -0.46.

2 The estimator: the learned pair energy as a kernel

The family is $K_\theta(a' \mid a) = \mathcal{N}(a'; a, \gamma)\exp(B(a')\theta)/Z_\theta(a)$ with B a linear basis for $\log g$ (Legendre polynomials on $[-A, A]$, or cubic B-splines), the constant function omitted (it is absorbed by Z), and Z the column normaliser under the grid's trapezoid weights, exactly as the MDN kernel is normalised. Parameters $(\log\gamma, \theta)$. In Jérôme's language, the pair energy is $W_\theta(a, a') = (a' - a)^2/2\gamma - \log g_\theta(a')$ and the transfer-matrix normaliser is $Z_\theta(a)$.

Lemma 5 (the M-step is concave in the tilt)**.** *With Ξ_{kj} the expected transition mass on $u_j \to u_k$ from the E-step and $N_j = \sum_k \Xi_{kj}$, $Q(\theta) = \langle \Xi, \log K_\theta \rangle = \sum_k (\sum_j \Xi_{kj}) B_k\theta - \sum_j N_j \log \sum_k w_k e^{B_k\theta + c_{kj}}$, linear minus a sum of log-sum-exp functions, hence concave in θ at fixed γ; its gradient is $\nabla_\theta Q = B^\top \sum_j \tilde{\Xi}_{\cdot j}$ with $\tilde{\Xi}_{kj} = \Xi_{kj} - N_j p_j(k)$, p_j the model's column distribution, and $\partial Q/\partial \log\gamma = \sum_{kj} \tilde{\Xi}_{kj}[(u_k - u_j)^2/2\gamma - \frac{1}{2}]$.*

Proposition 6 (what Boltzmann learning becomes for a kernel)**.** *By Fisher's identity the exact gradient of the noisy-data log-likelihood is $\langle \Xi, \nabla \log K_\theta \rangle$, which for this family is*

$$\nabla\mathcal{L} = \sum_j \Big[\sum_k \Xi_{kj}\, \nabla(-W_\theta(u_j, u_k)) - N_j\, \mathbb{E}_{K_\theta(\cdot \mid u_j)}\big[\nabla(-W_\theta(u_j, \cdot))\big] \Big] :$$

the positive phase is the posterior pair expectation (as in the note's §16.2) and the negative phase is the kernel's own conditional *expectation at each source state, weighted by the posterior source mass. The undirected pair-energy*

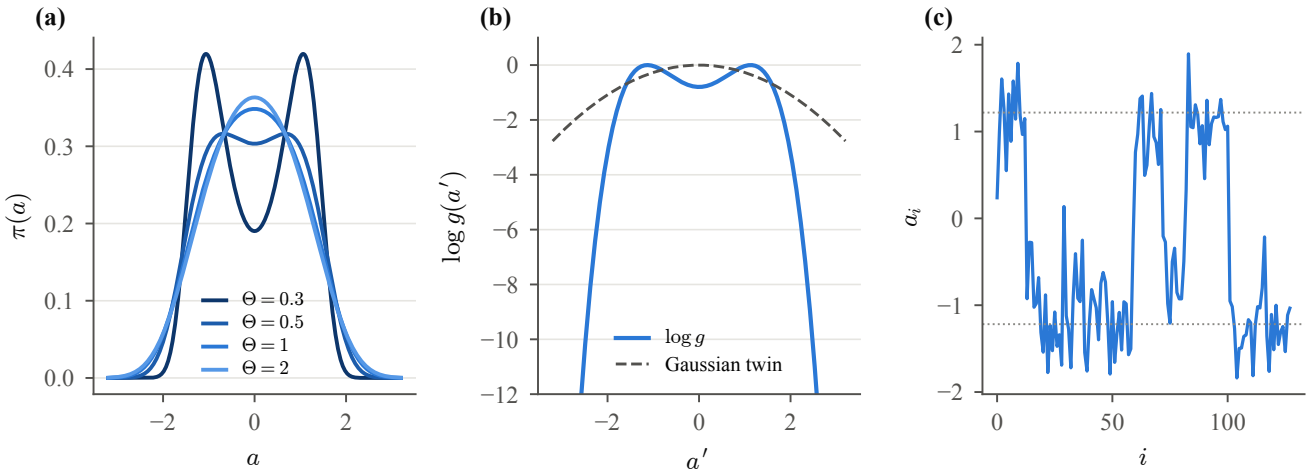


Figure 2: The chain at the temperature cells. (a) The standardised stationary law $\pi = \psi_0^2$: two wells at $\pm 1/s$ merging as Θ rises. (b) The arrival-point tilt $\log g$ at the base cell against the quadratic tilt of its Gaussian twin (Proposition 3): the truth is bimodal and super-Gaussian, the twin is a parabola. (c) One exact sample of 128 sites at the base cell; dotted lines at the wells. Domains of 5–25 sites separated by kinks.

model of the note's §16 has instead the chain's equilibrium *pair marginal as negative phase and a chain-wide* $\log Z_\theta$*. Both are exact maximum likelihood for their model; the directed form needs no* Z_θ*, its column normalisers are the ground-state transform (Proposition 2), and it contains the field's kernel exactly.*

Proposition 7 (every tilted walk is reversible)**.** *Let* $\gamma > 0$ *and* $g > 0$ *with* $Z_g(a) = \int \mathcal{N}(a'; a, \gamma) g(a')\, \mathrm{d}a' < \infty$*, and let* $K_g(a' \mid a) = \mathcal{N}(a'; a, \gamma) g(a')/Z_g(a)$*. Then* $\pi_g \propto g Z_g$ *is invariant and* K_g *is reversible with respect to it. Equivalently* K_g *is the ground-state transform of the symmetric operator* $\mathsf{T}_g(a, a') = u(a)\mathcal{N}(a'; a, \gamma)u(a')$ *with* $u = \sqrt{g/Z_g}$*, whose Perron eigenfunction is* $\psi = \sqrt{gZ_g}$ *with eigenvalue* 1*; so* K_g *is the Gibbs chain of the pair energy*

$$W(a, a') = \frac{(a' - a)^2}{2\gamma} + f(a) + f(a'), \qquad f = -\tfrac{1}{2}\log(g/Z_g), \tag{3}$$

and conversely every pair energy of this form is a tilted walk. For the field's (γ, g) *of* (2)*,* $f(a) = (h/2\Theta)V(sa) +$ const*.*

Proof. $\pi_g(a)K_g(a' \mid a) \propto g(a)g(a')\mathcal{N}(a'; a, \gamma)$ is symmetric in (a, a'): detailed balance, and invariance by integrating over a. For the second form, $(\mathsf{T}_g\psi)(a) = u(a)\int \mathcal{N}(a'; a, \gamma)u(a')\psi(a')\,\mathrm{d}a' = u(a)\int \mathcal{N}(a'; a, \gamma)g(a')\,\mathrm{d}a' = u(a)Z_g(a) = \psi(a)$ since $u\psi = g$; and $\mathsf{T}_g(a, a')\psi(a')/\psi(a) = \mathcal{N}(a'; a, \gamma)\, g(a')/Z_g(a)$ since $u\psi = g$ and $\psi/u = Z_g$. The converse is Proposition 2 with $D = e^{-f}$. For the field, $Z_g = \lambda_0\psi_0/D$ by (2), so $g/Z_g = D^2/\lambda_0$ and $f = -\log D + \text{const} = (h/2\Theta)V + \text{const}$. □

Numerically, on the $(401, 8)$ grid: π_g by the formula and by an eigensolve of K_g agree to 4×10^{-12}, detailed balance holds to 10^{-14}, and the site energy of the field's own kernel equals $(h/2\Theta)V(sa)$ to 2×10^{-10} (`tests/test_pair_energy_kernel.py`). The first question put to Jérôme and Marc in the morning version of this report (directed kernel or symmetric pair energy?) dissolves: no tilted walk is irreversible, and the symmetric form has the same single free function. What differs is the algorithm, per-column normalisers against the Perron eigenvalue.

Corollary 8 (a fitted kernel reads as physics)**.** *In standardised units* $\gamma = \Theta h/s^2$ *and* $f(a) = (h/2\Theta)V(sa) + \text{const}$*. A fit* $(\hat\gamma, \hat g)$ *of any kernel of this class gives*

$$\hat\Theta = \hat\gamma s^2/h, \qquad \hat V(\phi) = \frac{2\hat\Theta}{h}\,\hat f(\phi/s) + \text{const}, \qquad \hat f = -\tfrac{1}{2}\log\left(\hat g/Z_{\hat g}\right):$$

the temperature from the bond variance, the one parameter that enters no tilt; the potential, its barrier and its wells, from the site energy. Moreover the conditional mean step is exactly $\gamma(\log Z_g)'(a)$ *(exponential tilting), which for the field is* $\Theta h(\log\psi_0)' + (h^2/2)V'$ *in physical units, so the small-h limit is* $\mathrm{d}\phi = \Theta(\log\psi_0)'\,\mathrm{d}x + \sqrt{\Theta}\,\mathrm{d}B$*: the drift of the stationary field in the site index is the log-derivative of the ground state, not* $-V'$*.*

2.1 The physicist's parametrisation

Since $\log g = \log\psi_f - f$ is a transcendental function of the potential while f itself is a quartic, the parsimonious exactly specified model parametrises f. `PairEnergyKernel` (`src/markov_score/pair_energy_kernel.py`): $f(a) =$

$\sum_{m\le 4} c_m (a/2)^m$ and $\log\gamma$, five parameters, with the same fixed soft wall as an energy; K_f by one dense eigensolve of T_f per evaluation (15 ms on 401 points), invariant law ψ_f^2 in closed form; the M-step maximises $Q = \langle \Xi, \log K_f \rangle$ by L-BFGS-B with the analytic gradient of the Perron eigenvector from first-order perturbation theory in the complete discretised eigenbasis, $\partial_{c_m}\psi_0 = -\sum_{n\ge 1} \psi_n \frac{\lambda_0+\lambda_n}{\lambda_0-\lambda_n} \langle \psi_n \mid B_m \psi_0 \rangle$, checked against finite differences and Fisher's identity to 10^{-4}. Q is not concave in c, so a step is accepted only if Q did not decrease, as for every other kernel. With the field's own (γ, c) the kernel reproduces the chain's to 2×10^{-3} (the coarse eigensolve against the fine one), and from 1024 clean chains at $(\Theta, h) = (0.3, 1)$ it reads $\hat{\Theta}$ within 5 % and the barrier within 10 %. This is literally the pair energy W_θ of the email, its Boltzmann learning written as a kernel. It joins E33 as a fifth arm, `em_pair_quartic`, under amendment 1 of the registry (predictions P7 and P8, declared before any E33 kernel result existed), run as its own shards so the four registered arms' tables are never touched. In the smoke run at $M = 32$ it already sits at 0.13 against the polynomial tilt's 0.25: five parameters have the smaller variance.

Implementation (`src/markov_score/tilted_kernel.py`): L-BFGS-B on $(\log\gamma, \theta)$ with the analytic gradient, warm-started, accepted only if Q did not decrease (a rejected step returns the receiver's parameters in a new object, so the checkpoint ladder stays honest); Fisher's identity checked against finite differences of the evidence to 10^{-5}; the Gaussian embedding recovered on a Gaussian chain, and the field's kernel recovered from clean data with the spline basis (`tests/test_tilted_kernel.py`). Two details earned by the first smoke run:

- **The tilt carries a fixed soft wall** $-(a/5)^{12}$ on the $(401, 8)$ grid. The data occupy $|a| < 4$, so the basis beyond that is constrained only through far columns' normalisers, and a free fit can raise the tilt at the grid edge and create a metastable pocket there, invisible to the likelihood but the place where the kernel's invariant law then puts mass. The wall is 0.002 at $|a| = 3$, 0.07 at 4 and 280 at 8, more than any coefficient within the optimiser's bounds can undo; the field's true tilt is already $-0.2a^4$ there.
- **The initial law is the kernel's own invariant density**, by a direct eigensolve of the 401×401 operator (`em.fit_em` takes a callable `log_mu`): the model is a stationary chain, as the data are. The M-step maximises the edge term; any evidence decrease is recorded.

3 What does not extend: Gaussian closure of the messages

The paper's Proposition 1 identifies moment-projected sum-product with the LMMSE of the covariance twin, and its proof uses the linear transition. On the field chain the projected recursion is not a well-posed estimator, for a reason that was found by the diagnostic, not anticipated.

Proposition 9 (the backward message saturates)**.** *Let* $\log g(a') \le -c\,a'^4 + C$ *(a super-Gaussian tilt,* $c > 0$*). Then* $K(\cdot \mid a)$ *concentrates, as* $|a| \to \infty$*, near* $a'_*(a)$ *with* $|a'_*| \sim (|a|/4c\gamma)^{1/3}$*, and the backward message* $\overleftarrow{m}(a) = \int K(a' \mid a)\ell(a')\,\mathrm{d}a' = \mathbb{E}[\ell(a_{i+1}) \mid a_i = a]$ *is bounded and decays only as* $\exp(-c'|a|^{2/3})$*.*

Sketch. The exponent $-(a'-a)^2/2\gamma - ca'^4$ is maximised where $(a-a')/\gamma = 4ca'^3$, i.e. $a'_* \approx (a/4c\gamma)^{1/3}$ for large $|a|$, with curvature $\sim 12ca'^2_*$, so $K(\cdot \mid a)$ is a sharpening Gaussian at a'_* and $\overleftarrow{m}(a) \approx \ell(a'_*(a))$; for the channel's $\ell(a') \propto \exp[-(x - e^{-t}a')^2/2\Delta_t]$ this is $\exp[-(x - e^{-t}a'_*)^2/2\Delta_t]$. □

So on a finite grid $\overleftarrow{m}$ is nearly constant, and the Gaussian with its moments is close to flat: the projection discards the backward information. Measured at $t = 0.47$ on the $(401, 8)$ grid, 28 % of a projected message's mass sits in the outer 5 % of the grid; over E32's cells and levels the maximum is 42 % at the base cell and 98–100 % at every other cell (Table 2). On grids wide enough for the Gaussian and Laplace chains' backward messages (half-width 32 at $t = 1$) the projected recursion reproduces the LMMSE to 10^{-8} and 10^{-5}, so this is a property of the field, not of the implementation (`tests/test_bp_projected.py`).

Consequently the Gaussian-closure arm of the field chain is its covariance twin, the LMMSE of the true covariance, by the identification Proposition 1 proves on linear chains; the projected recursion is kept as the diagnostic and its edge mass is recorded. Jérôme's expectation that closure "fails badly" by smearing across domain walls is then measured by the linear floor, which is what the floor has always measured, and it does: 0.269 at the base cell. For the paper this is a scope sentence on Proposition 1 and one sentence in the field section.

4 E32: the laboratory before any learning

Deterministic, registered and predeclared, run on the laptop at commit `9b1c940`, frozen (`outputs/E32`, 9 files). Three predictions were registered first.

P1, the grid: confirmed far inside its bound. The $(401, 8)$ reference score agrees with the $(1601, 8)$ one to 10^{-12} relative or better on every cell and level, against a declared tolerance of 10^{-3}. The field kernel is smooth, so the trapezoid rule is spectrally accurate here, unlike the Laplace chain's kink. The same holds for the eigensolve: halving the fine spacing or widening the domain moves λ_0, ρ_1 and s by less than 10^{-10}.

P2, the projection: confirmed. The edge mass exceeds 10 % at some level of every cell (Table 2); the closure arm is the covariance twin.

cell	linear floor	isotropic	projected (diag.)	edge mass	grid error
$\Theta = 0.3$, $h = 1$, $L = 32$	0.269	0.674	0.262	0.42	9.6×10^{-15}
$\Theta = 0.5$, h^*, $L = 32$	0.115	0.658	0.278	1.00	1.3×10^{-12}
$\Theta = 1$, h^*, $L = 32$	0.058	0.657	0.208	0.99	1.3×10^{-15}
$\Theta = 2$, h^*, $L = 32$	0.041	0.656	0.169	0.98	1.6×10^{-15}
$\Theta = 0.3$, $h = 0.5$, $L = 64$	0.162	0.714	0.291	1.00	2.3×10^{-15}
$\Theta = 0.3$, $h = 0.25$, $L = 128$	0.098	0.750	0.529	1.00	1.1×10^{-15}

Table 2: E32 ladder, pooled relative score error over the twelve-level schedule (mean of two replicates of 512 chains), the projected recursion's largest edge mass over levels, and the worst relative error of the $(401, 8)$ reference score against $(1601, 8)$ over levels. For scale: the Laplace laboratory's floor at $\rho = 0.85$ is 0.168 and E14b's bimodal chain's 0.082.

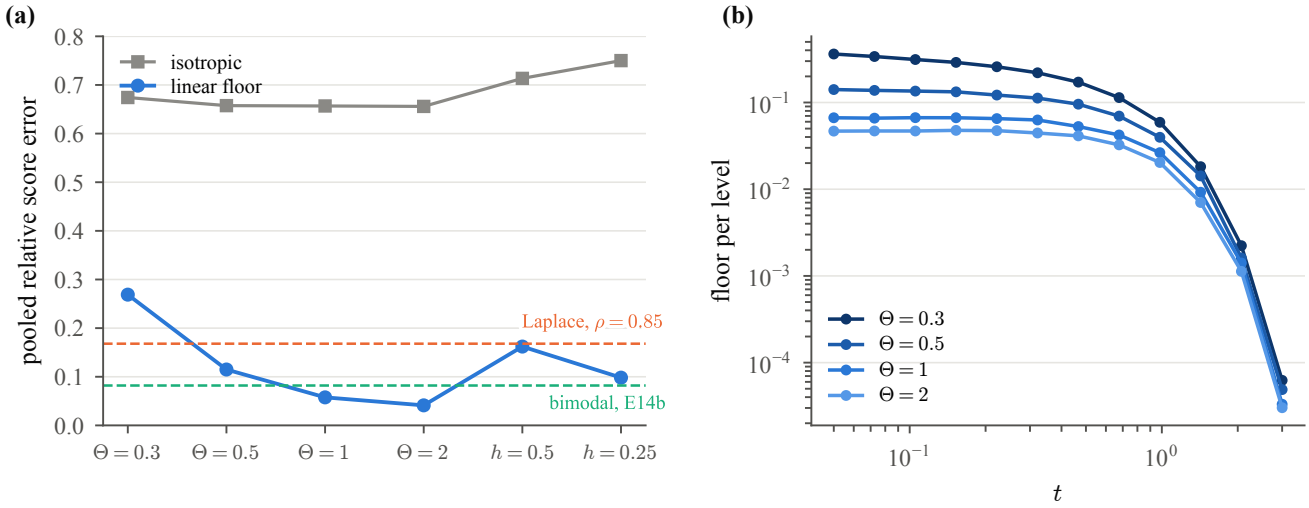


Figure 3: (a) The linear floor and the isotropic map across the six cells; dashed references are the laboratory's Laplace floor and E14b's bimodal floor at the same correlation. (b) The floor per level at the four temperature cells: the field's second-order deficit lives at low t and is largest where the wells are deepest.

P3, Theorem 2: consistent, not reached within the schedule. The ratio rises monotonically through t_* and reads 0.85, 0.87, 0.73, 0.94 at $t = 3$ on the four cells (Fig. 4). The linear chains reached the constant at $t = 4$ with 8192 chains (E18); this run stops at $t = 3$ with 512, and the ratio at $t = 2$ already moves by 0.1 between the two replicates. The constants are large, $C_4 = \frac{1}{6}\|\kappa_4\|_F^2 = 3882, 2350, 1261, 838$ at $\Theta = 0.3, 0.5, 1, 2$ for $L = 32$, which is why the gap is still 10^{-2}–10^{-1} of the score energy at $t = 1$. Extending the schedule to $t = 4$ with 8192 chains is cheap and is the one addition E32 needs.

The floor itself. At the base cell the linear floor is 0.269, well above the Laplace laboratory's 0.168 at the same correlation and more than three times E14b's bimodal chain's 0.082 (kurtosis -0.72): the field at $\Theta = 0.3$ carries more structure beyond second order than any chain the paper has used, and the reference score has that much room above the floor. The floor falls monotonically along the temperature sweep ($0.269 \to 0.041$), which is E33's P6, and along the refinement cells ($0.269 \to 0.162 \to 0.098$), because at finer h the per-step kernel is closer to Gaussian while the chain is more correlated.

4.1 The temperature sweep on the note's own axis

The registered sweep sits on the iso-correlation line, which holds ξ in sites fixed by construction; the consolidated note (§24.6, its figure 5) reads the temperature at fixed spacing, where lowering Θ lengthens the domains and sharpens the bimodality together. Amendment 1 of E32 adds that axis: eight cells at $h = 1$, $L = 32$, $\Theta \in \{0.2, 0.3, 0.4, 0.5, 0.7, 1, 1.5, 2\}$, the oracle ladder and E34's physical statistics on 4096 exact chains, two reps, frozen the same afternoon (Figure 5, Table 3). Two predictions were registered before it ran. P4, that the linear floor falls monotonically with Θ at fixed h and exceeds the base cell's 0.269 at $\Theta = 0.2$, is confirmed: 0.296 at $\Theta = 0.2$, then $0.269, 0.241, 0.225, 0.205, 0.194, 0.186, 0.184$. P5, as written, said that the coarse kink density and the bimodality coefficient both fall with Θ; the bimodality does ($0.73 \to 0.45$) and so does the domain length ($9.0 \to 5.3$ in physical units), but the kink density *rises* ($0.016 \to 0.166$ per unit length), which is the physics (more kinks at higher temperature, roughly Arrhenius in $1/\Theta$) and a sign error in the prediction's sentence. A mis-stated prediction counts as falsified; the registry says so.

The two axes answer different questions, and the comparison is the point. On the iso line the floor falls from

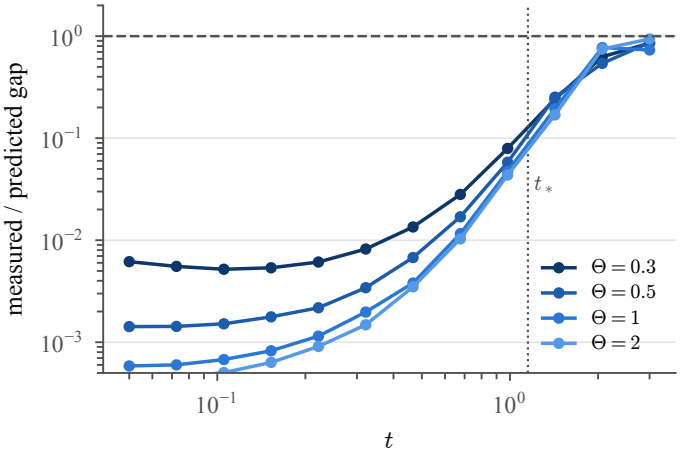


Figure 4: Theorem 2 on the second laboratory: $\mathbb{E}\|S - S^{\mathrm{G}}\|^2$ from grid BP against the linear floor, divided by $\frac{1}{6}\|\kappa_4\|_F^2 e^{-8t}$ with κ_4 from transfer-matrix four-point functions, at the four temperature cells; $t_* = \frac{1}{2}\log\lambda_{\max}(\Sigma_0) = 1.15$.

Table 3: The fixed-spacing sweep ($h = 1$, $L = 32$; mean of two reps). ξ in sites, domain length and kink density in physical units after merging runs shorter than the kink width, b the bimodality coefficient, κ_4 the marginal excess kurtosis, and the pooled score error of the two oracle arms.

Θ	ξ	$\bar{\ell}_{\mathrm{dom}}$	ρ_{kink}	b	κ_4	closure	isotropic
0.2	19.5	8.97	0.016	0.73	−1.64	0.296	0.751
0.3	6.08	7.79	0.066	0.61	−1.35	0.269	0.674
0.4	3.66	7.04	0.099	0.54	−1.16	0.241	0.610
0.5	2.75	6.47	0.119	0.51	−1.04	0.225	0.562
0.7	1.98	5.95	0.138	0.48	−0.92	0.205	0.497
1	1.53	5.66	0.151	0.46	−0.85	0.194	0.438
1.5	1.21	5.40	0.161	0.46	−0.80	0.186	0.381
2	1.06	5.31	0.166	0.45	−0.78	0.184	0.347

0.27 to 0.04 between $\Theta = 0.3$ and 2: at fixed second-order structure, what the closure loses is the non-Gaussian shape, and that shape fades as the marginal approaches a Gaussian. At fixed spacing the floor falls only from 0.30 to 0.18: a hot cell has $\xi \approx 1$ site, so there is little sequence structure to lose, but its marginal still has kurtosis -0.78, and the closure cannot denoise a non-Gaussian marginal even site by site. The note's expectation that "all methods come closer at high Θ" is therefore true of the structured arms against each other, not of the linear closure against the reference score, which keeps a floor of 0.18 at $\Theta = 2$ on this axis. For the learned arms the fixed-h cells are not in E33; whether to add them is question 3 to Jérôme and Marc.

4.2 The subsampling protocol: one field at three strides

The note's §21.1 recommends generating at fine resolution and subsampling; the registered refinement cells are instead three lattice models. E36 (registered and predeclared this evening) observes one fine field ($\Theta = 0.3$, $h = 0.25$, 128 sites) every s-th site, $s \in \{1, 2, 4\}$: the observed chain is exactly Markov with kernel K^s (Chapman–Kolmogorov on the grid, same invariant law, λ_1/λ_0 to the power s), so the refinement question is asked of one physical object. Two of its three predictions are already read. P1, that the coarse kink density and the coarse domain length agree across strides within 10 %, holds: $0.068, 0.073, 0.070$ per unit length and $8.1, 7.6, 7.8$ physical units (7.5 % and 6.7 % spread); the raw kink density per bond does not ($0.061, 0.087, 0.130$), N-31 again. P2, that the stride-4 floor is within 15 % of the $h = 1$ lattice cell's, holds at 12.8 %: 0.235 against 0.269. Across the three spacings the strided floors are $0.098, 0.158, 0.235$ against the lattice models' $0.098, 0.162, 0.269$: the two protocols agree at the fine end and part by 13 % at the coarse one, where the $h = 1$ lattice model is the rougher object (marginal kurtosis -1.35 against the fine field's -1.22). The third prediction concerns learning: for $s \geq 2$ the kernel K^s is no longer a tilted walk (the Gaussian bond does not survive marginalising the intermediate sites), so the gradient-class kernels are misspecified on the subsampled chain while a state-dependent mixture is not.

Table 4 reads P3. Its first two parts hold: at stride 1 the gradient-class kernels are exactly specified (risk 0.0013 to 0.0021), and at stride 4 their risk is 18 to 30 times larger on both seeds. The third part, that the neural kernel beats the site energy at stride 4, holds on one seed of two (0.023 against 0.040, then 0.093 against 0.041); the neural kernel selected its last checkpoint in all six cells, so its numbers are limited by the iteration budget and the clause is unresolved rather than refuted. Two observations were not registered and are the more useful ones. The mixture kernel sits *on* the linear floor at every stride (ratio 1.007 to 1.026): on the field, the paper's

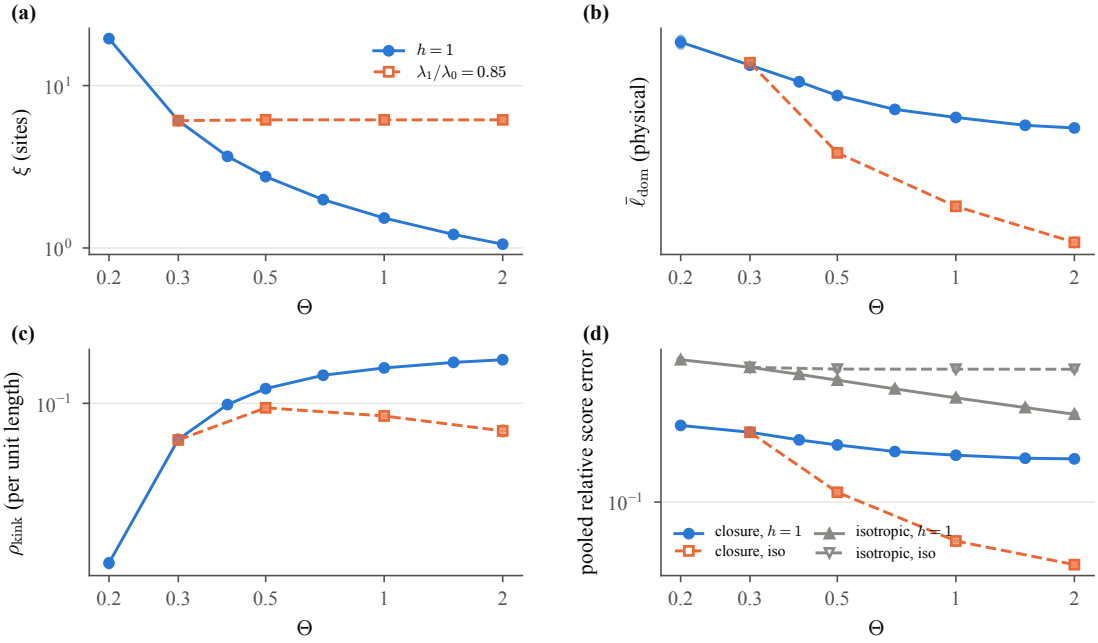


Figure 5: The temperature sweep on both axes: fixed spacing $h = 1$ (filled, solid) and matched correlation $\lambda_1/\lambda_0 = 0.85$ (open, dashed). (a) Correlation length in sites from the spectrum. (b) Coarse mean domain length in physical units. (c) Coarse kink density per unit length. (d) Pooled relative score error of the Gaussian closure and of the isotropic map. Two reps per cell.

Table 4: E36, clean-data fits on 2048 strided chains (seeds 0 and 1), scored against grid BP with the true K^s. "Floor" is the linear floor of the same strided chain.

stride s	site energy (5)	spline tilt (25)	mixture (25)	neural kernel (1,516)	floor
1	0.0013, 0.0017	0.0018, 0.0021	0.100, 0.100	0.016, 0.013	0.098
2	0.011, 0.012	0.010, 0.011	0.159, 0.159	0.013, 0.034	0.158
4	0.040, 0.041	0.036, 0.036	0.238, 0.237	0.023, 0.093	0.235

estimator is the Gaussian closure even with clean data, because its drift is linear. And subsampling costs the gradient-class kernels a factor of 20 to 30 in risk but not their advantage: at stride 4 they are still at 15 to 17 % of the floor.

5 E33 and E34: the design

5.1 E33, learning through noise

On the six cells, under the registered protocol (sizes $M \in \{32, 128, 512, 2048\}$, seeds 0–15, bundles keyed by tag, seed, cell and M, one noised view per sequence spread over the twelve levels, validation 1024 and test 2048 chains per seed):

- kernel arms through grid BP, the same ladder to 1200 for every kernel, the kernel's own invariant law as initial law, both of E14's selection rules (held-out BP risk; held-out one-view evidence): `em_mixture` ($C = 8$, 25 parameters, misspecified), `em_tilted_poly` (Legendre degree 6, 7), `em_tilted_spline` (24 B-splines, 25), `em_mdn` (E25/E30's neural kernel, 1,516), and, by amendment 1, `em_pair_quartic` (the site energy itself, 5; §2.1);
- the window head and the transformer, re-screened per cell on development seeds and trained two ways, imported unchanged from E14: implicit score matching on the one-view bundle (the matched interface) and denoising score matching on clean targets with one fixed corruption (the network-favourable bracket), at $M \in \{2048, 128\}$;
- per (cell, seed): the linear floor, the isotropic map, the projected diagnostic.

Registered predictions. P1 the mixture keeps an asymptotic bias on every cell (above the spline in 16/16 seeds at $M = 2048$, the gap not closing from 512 to 2048). P2 the learned pair energy beats the mixture at every $M \geq 128$ on every cell, the 7-parameter tilt within 20 % of the 25-parameter one at $M = 2048$. P3 the MDN improves relative to the mixture compared with E30 (mdn/mixture below 1.5 at $M = 2048$): a state-dependent mean is what this chain needs. P4 no network reaches the spline at any M on any cell under either interface. P5

the window head's deficit grows along the refinement cells while the spline's does not. P6 the floor at the base cell is above E14b's 0.082 and falls along the sweep. Amendment 1 (afternoon, before any kernel result): P7 the five-parameter site energy is within 10 % of the spline at $M = 2048$ on every cell and below it at $M = 32$ and 128 (16/16 paired); P8 $\hat{\Theta}$ is within 10 % of Θ at $M = 2048$ for both the quartic and the spline (16/16), and the quartic's barrier within 20 % of 1/4 on the $\Theta \leq 1$ cells.

Oracles, in. Job 678084 completed after 2 h 04 and was fetched: 288 rows, sixteen seeds on all six cells. P6 is *confirmed* on the test bundles: the linear floor is 0.269 at the base cell (E14b's bimodal floor at the same correlation is 0.082) and falls monotonically along the sweep, 0.269, 0.114, 0.059, 0.041 at $\Theta = 0.3, 0.5, 1, 2$, with a spread over seeds below 0.001; on the refinement cells 0.164 and 0.098 at $h = 0.5, 0.25$. The projected diagnostic carries 38–43 % of its mass at the grid edge on the base cell and 98–100 % on every other cell, which is the measured form of §3.

Jobs, deployment `9b1c940-20261001T084745Z`, submitted 08:55 UTC: kernels 678087–678090 (`defq`, four seed shards, each its own directory); screen 678091 then nets 678092/678094 (`long_gpuh200`, one H200 each; the A100 partition was in maintenance); all still queued at the time of writing. The kernel shards are estimated at about eight hours each on 24 workers, the $L = 128$ cells at $M = 2048$ being the long pole; the network jobs at one to two days. The quartic arm's four shards are submitted from the afternoon's deployment (§2.1).

5.2 E34, the physics and the clean-data interface

The consolidated note (§24.5, §16.1) asks for physical statistics beside the score error and for the clean-data baseline. E34 supplies them on E33's streams, so every row pairs with E33's: `physics_truth` (kink density per bond and per unit length, interior domain lengths, $C(r)$ against the spectral sum, kurtosis against the exact one, barrier mass, on 4096 exact chains per cell; laptop); `clean` (the four kernel arms fitted by clean-data maximum likelihood on E33's training chains, scored on E33's test bundles: the price of the channel per family; one cluster job); `physics_learned` (ancestral samples from every fitted E33 kernel at $M = 2048$, started from its own invariant law, the same statistics against the truth; after E33). Four predictions registered; the quick smoke already shows the direction of two of them: on clean data at $M = 32$ the mixture sits at 0.27 while the polynomial tilt reaches 0.04, and samples from a fitted mixture overshoot the truth's kink density (0.21 against 0.13 per bond) while the spline's match it.

5.3 First readings, from a laptop preview

While the cluster shards queue, a preview of the kernel arms runs on the laptop: the base cell and the $\Theta = 2$ cell, seeds 0 and 1, $M \in \{128, 512\}$, under the registered code path into separate directories (not confirmatory, two seeds). The five-parameter site energy finished first. Figure 6 and Table 5 are Corollary 8 applied to its risk-selected fits at $M = 512$, i.e. to kernels learned *through the noise* from 512 sequences of 32 sites each seen at one level.

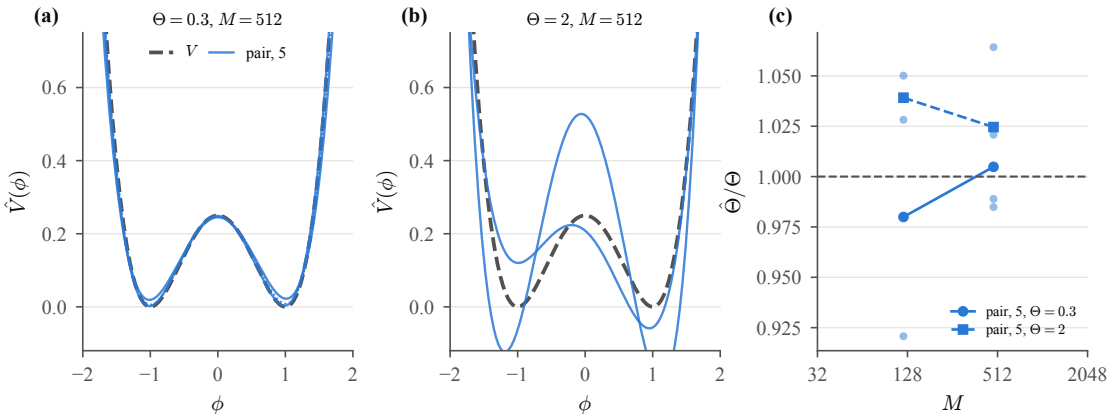


Figure 6: The potential and the temperature read from kernels learned through noise (preview, two seeds). (a) $\hat{V}(\phi)$ from `em_pair_quartic` at the base cell, $M = 512$, each seed a thin line, the true double well dashed, a clean-data fit dotted. (b) The same at $\Theta = 2$. (c) $\hat{\Theta}/\Theta$ against M. Regenerated by `tools/make_field_potential_figs.py`; the confirmatory shards replace the preview when they land.

Three readings. First, the temperature is identified to about a percent and the double well to 0.016 RMS from noised data at $M = 512$, with a score risk of 0.021 against the Gaussian closure's 0.269 on the same cell: a factor thirteen, by a kernel with five parameters. At $M = 128$ the temperature is still read within 2 % on average (spread 8 %) and the barrier within 3 %; clean data at $M = 2048$ read all three to 0.1–0.5 %, so the channel costs a factor of a few in risk and a few percent in the physics at these sizes, not the identification. The same fitted kernels, sampled ancestrally from their own invariant law (E34 `physics_learned` on the preview), reproduce the

Table 5: Readings from the risk-selected fits (preview; mean over the seeds, spread in brackets). Truth: $\hat{\Theta}/\Theta = 1$, barrier 1/4, wells at ± 1. "RMS" is the root-mean-square deviation of $\hat{V}$ from V on $|\phi| \leq 1.6$ after removing the constant.

cell	arm, data	M	$\hat{\Theta}/\Theta$	barrier	well	RMS	risk
$\Theta = 0.3$	site energy, noisy	128	0.980 (0.084)	0.256 (0.033)	1.02	0.019	0.035
$\Theta = 0.3$	site energy, noisy	512	1.005 (0.023)	0.234 (0.016)	1.00	0.016	0.021
$\Theta = 0.3$	site energy, clean	512	0.985 (0.010)	0.236 (0.002)	1.00	0.009	0.010
$\Theta = 0.3$	site energy, clean	2048	0.999 (0.005)	0.250 (0.000)	1.00	0.003	0.005
$\Theta = 0.3$	spline tilt, clean	2048	0.999 (0.005)	0.250 (0.003)	1.00	0.004	0.006
$\Theta = 2$	site energy, noisy	512	1.025 (0.056)	0.50 (0.33)	1.04	0.13	0.019
$\Theta = 2$	site energy, clean	2048	1.003 (0.004)	0.29 (0.02)	1.04	0.03	0.004

truth's kink density per bond (0.123 and 0.130 against 0.123 and 0.120 on the two seeds) and its mean domain length (5.1 and 4.9 sites against 5.1 and 5.0): the generated-sample quality the note asks for, in the note's own statistics. Second, at $\Theta = 2$ the temperature is still read to 0.3–2.5 % but the barrier is not: 0.50 ± 0.33 through noise, 0.29 ± 0.02 even from clean data. At that temperature the thermal energy is eight times the barrier and the marginal is near-Gaussian, so the data carry almost no information about the shape of V between the wells; the risk is nonetheless 0.019 against the closure's 0.041, because what the score needs there is the bond and the curvature, which are identified. P8 was written to expect this. Third, the clean-data fits read everything to 0.5 % at the base cell: the price of the channel at $M = 2048$ is a factor of a few in risk, not in the physics.

One negative result came with the preview (N-33 in `docs/NEGATIVE_RESULTS.md`, corrected on 2 October): the *polynomial* tilt sometimes fails badly, with risks of 0.18 to 1.79 in five of its twenty preview fits, clean and noisy. The cause is not where the fit puts its transitions but where it puts its *invariant law*. The likelihood constrains $\log g$ only where data arrive ($|a| \lesssim 3.5$); beyond that, a global basis can raise the tilt and create a pocket, and in every failed fit the kernel's own invariant law sits entirely in that pocket. Since the estimator uses that law for the first site, every posterior is dragged outside the data. In the other fifteen Legendre fits, and in all fifty-two spline and site-energy fits (the previews and E36), the mass beyond $|a| = 3.5$ is at most 1.1×10^{-4}. That mass, not the coefficient bound first suspected, is the diagnostic: E33's amendment 2 reports the number of such "parked" fits beside every verdict, before any confirmatory row exists. The role of the parsimonious exactly specified kernel passes to the site energy, which confines by construction.

6 Confirmatory results

Every registered run of the field laboratory has landed. E33: the four kernel arms (3072 rows, jobs 681261–681265, rerun under amendment 3), the site energy of amendment 1 (768 rows, jobs 678260–678263), the oracles and the two networks under both interfaces (768 rows, jobs 678092 and 678094); six cells, sixteen seeds, sizes $M \in \{32, 128, 512, 2048\}$ (the networks at 128 and 2048). E34: the clean-data arm (1536 rows, job 678166) and `physics_learned` on the laptop. E35 on the laptop. Each was read by the registered analysis (`tools/make_field_numbers.py`, with the parked count of amendment 2 and the failure rule of amendment 3 beside every verdict) and frozen. Every table below is generated from the frozen files, by `tools/make_field_confirmatory_tables.py`; none is typed by hand.

6.1 The scoreboard

Table 6 is the answer to the email in one table. The five-parameter site energy is the best estimator on every cell at every size; at the base cell it reaches a thirtieth of the closure's risk from 2048 sequences. The paper's mixture kernel converges *onto* the linear floor from above, 1.00 to 1.02 times it at $M = 2048$ on every cell: on this chain its linear drift learns exactly the second-order structure and nothing else, and more data only remove its variance. The MDN, whose mean is state-dependent, beats the mixture everywhere but stays two to six times above the spline at $M = 2048$. The networks are four to twenty-seven times above the spline at every cell and size, and at $\Theta \geq 0.5$ they do not reach even the linear floor at $M = 2048$. The registered verdicts:

- **P1 confirmed**: mixture/spline above one in 16/16 seeds on every cell (2.5 to 18 at $M = 2048$), the gap widening, not closing, from 512 to 2048 (by 1.5–2.2).
- **P2 falsified as written**: the spline beats the mixture 16/16 at every $M \geq 128$ on the four cells with $\Theta \leq 0.5$, not at $\Theta = 1$, $M = 128$ (12/16) or $\Theta = 2$, $M = 128$ and 512 (4/16, 15/16), where the near-Gaussian marginal keeps the linear drift competitive; the Legendre tilt fails on the finest lattice. Its second clause, Legendre within 20 % of the spline, holds only at the base cell: elsewhere the Legendre is 29–43 % better ($\Theta \geq 0.5$) or 53 % worse ($h = 0.5$).
- **P3 confirmed**: MDN/mixture 0.23–0.82 at $M = 2048$ on every cell.

Table 6: E33, score risk over the linear floor (the covariance twin, E33's oracles), mean over sixteen seeds, through one noised view per sequence. Below one beats the Gaussian closure. Kernel arms by EM through grid BP, parameter counts in brackets; networks under the two interfaces. A superscript counts failed fits (amendment 3), excluded from the mean; "failed": all sixteen.

M	arm	$\Theta = 0.3$	$\Theta = 0.5$	$\Theta = 1$	$\Theta = 2$	$h = 0.5$	$h = 0.25$
128	site energy (5)	0.139	0.269	0.534	0.630	0.136	0.130
	spline tilt (25)	0.218	0.454	1.111	1.478	0.231	0.235
	Legendre tilt (7)	0.394	0.354	0.647	0.816	0.330	0.387^{6}
	MDN	0.468	0.900	1.724	2.457	0.476	0.585
	mixture (paper)	1.018	1.039	1.187	1.279	1.016	1.017
	window, ISM	1.026	4.396	6.264	8.735	1.495	2.237
	transformer, ISM	3.284	9.752	8.509	14.487	3.591	4.729
	window, DSM	0.829	2.339	4.149	5.786	1.125	1.580
	transformer, DSM	1.787	4.359	8.034	11.571	1.887	2.129
2048	site energy (5)	0.033	0.072	0.141	0.169	0.040	0.032
	spline tilt (25)	0.060	0.134	0.280	0.429	0.073	0.068
	Legendre tilt (7)	0.074	0.092	0.165	0.243	0.111	failed
	MDN	0.228	0.422	0.737	0.837	0.310	0.420
	mixture (paper)	1.004	1.004	1.014	1.021	1.010	1.016
	window, ISM	0.552	1.284	2.457	3.559	0.776	1.196
	transformer, ISM	1.354	2.706	2.440	3.237	1.368	1.020
	window, DSM	0.484	1.174	2.287	3.334	0.701	1.030
	transformer, DSM	0.533	1.225	2.156	3.043	0.622	0.780

- **P4 confirmed**: no network reaches the spline, 16/16 everywhere.
- **P5 falsified**: the window head's risk at $M = 2048$ *falls* along the refinement (0.149, 0.127, 0.118); what grows is its ratio to the spline (9, 11, 18) and to the floor (0.55, 0.78, 1.20): on the finest lattice the window head is worse than the Gaussian closure.
- **P6 confirmed** (§5).
- **P7 falsified as written, favourably**: the site energy is not within 10 % of the spline at $M = 2048$ but 43–59 % below it, and below it at $M = 32$ and 128 on every cell (one seed short at the base cell, $M = 32$).
- **P8**: the temperature within 10 % for both arms in all 192 fits; the barrier in 78 of 80 (§6.2).
- **E34**: P3 and P4 confirmed, P2 falsified as written (§6.5).

6.2 The site energy through the noise (E33, P8)

Table 7: `em_pair_quartic`, risk-selected fits through one noised view per sequence. Middle columns: $M = 2048$, mean over sixteen seeds with the seed spread in brackets; "band" counts barriers within 20 % of 1/4 (registered for $\Theta \leq 1$). Right: the score risk as a fraction of the linear floor (E33's oracles) at each M.

	$M = 2048$			risk / linear floor			
cell	$\hat{\Theta}/\Theta$	barrier	band	32	128	512	2048
$\Theta = 0.3$, $h = 1$	1.000 (0.023)	0.247 (0.011)	16/16	0.272	0.139	0.067	0.033
$\Theta = 0.5$	1.005 (0.012)	0.246 (0.015)	16/16	0.657	0.269	0.159	0.072
$\Theta = 1$	1.005 (0.021)	0.248 (0.036)	14/16	1.367	0.534	0.278	0.141
$\Theta = 2$	1.000 (0.013)	0.277 (0.083)	–	1.429	0.630	0.414	0.169
$\Theta = 0.3$, $h = 0.5$	1.007 (0.008)	0.254 (0.007)	16/16	0.277	0.136	0.063	0.040
$\Theta = 0.3$, $h = 0.25$	1.001 (0.003)	0.250 (0.007)	16/16	0.237	0.130	0.052	0.032

The temperature clause of P8 holds in all 96 site-energy fits and all 96 spline fits ($\hat{\Theta}/\Theta$ means 0.999–1.009). The barrier clause fails as written on one cell: at $\Theta = 1$ two of sixteen barriers (0.163 and 0.313) leave the 20 % band, although the mean is 0.248. At $\Theta = 1$ the barrier is a quarter of the thermal energy and its reading spreads three to five times more than at $\Theta \leq 0.5$; the registered exclusion stopped at $\Theta = 2$, where the reading is indeed unresolved (0.277 ± 0.083). At $\Theta = 0.3$ five parameters learned from 32 noised sequences already carry a quarter of the closure's risk; at $\Theta \geq 1$, where the marginal is nearly Gaussian, 32 sequences are not enough to beat the closure and 128 are.

6.3 Generated samples (E34, P4)

P4 holds on every cell: samples from the spline kernels reproduce the kink density and the domain length within 2 % on every cell mean, and samples from the mixture create 1.08 to 1.70 times too many kinks, with domains up

Table 8: Generated over true kink density and mean domain length, 2048 chains sampled ancestrally from each risk-selected fit at $M = 2048$ (seeds 0–3), mean over the seeds. P4 names the spline (within 15 %) and the mixture (overshoots the kinks); the other rows are unregistered.

arm	$\Theta = 0.3$	$\Theta = 0.5$	$\Theta = 1$	$\Theta = 2$	$h = 0.5$	$h = 0.25$
site energy (5), kinks	1.003	1.008	0.978	1.004	1.016	1.002
site energy (5), domain	1.003	1.001	1.015	0.999	0.996	0.986
spline tilt (25), kinks	1.008	1.008	1.001	1.011	1.006	1.011
spline tilt (25), domain	1.001	1.001	1.008	0.994	0.997	0.982
Legendre tilt (7), kinks	1.016	1.010	0.995	1.004	0.969	failed
Legendre tilt (7), domain	0.983	0.996	1.006	1.012	0.987	failed
MDN, kinks	1.004	1.009	0.992	1.009	1.018	1.050
MDN, domain	0.999	0.996	1.011	0.988	0.980	0.971
mixture (paper), kinks	1.529	1.215	1.115	1.083	1.624	1.704
mixture (paper), domain	0.766	0.904	0.958	0.961	0.722	0.692

to 31 % too short, worst where the wells are deepest. A linear drift cannot hold a domain. The site energy, the MDN and (where it fitted) the Legendre tilt reproduce both statistics within 5 % on every cell mean; for the site energy, fit by fit, the kink density is within 5.3 % and the domain length within 2.9 % in all 24 fits, and the coarse kinks, bimodality and variance within 13.3 %, 2.9 % and 4.0 % (`outputs/E34_learned_pair`).

The note asks for the domain-length *distribution*, not only its mean (its section 24.5). E37 compares the whole distribution of coarse domain lengths on the same samples: for the spline and the site energy the two-sample Kolmogorov–Smirnov test never rejects (p-values 0.36–0.88 on every cell) and their 10, 50 and 90 % quantiles sit within 10 % of the truth's, but for one 10 % quantile at $h = 0.5$ (1.104, one lattice step); the mixture's domains are 14–18 % short at the median on the three $\Theta = 0.3$ cells, its KS distance four to five times that between two independent exact samples. Both E37 predictions fail as written for reasons of calibration, not of the samples: a KS band of 0.05 sat below the sampling noise at $\Theta \geq 1$, where a sample holds only 350 to 700 domains, and the mixture's tail deficit (12–13 % at the 90 % quantile) was overstated at 15 %.

6.4 Refinement and runtime (the note's figure 6)

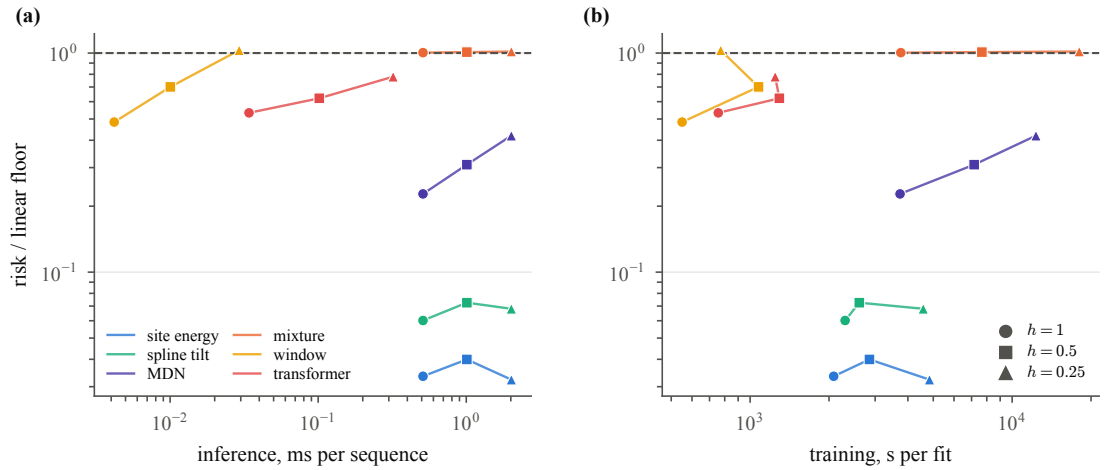


Figure 7: Accuracy against cost on the three refinement cells (circle $h = 1$, square $h = 0.5$, triangle $h = 0.25$), $M = 2048$, sixteen seeds; the dashed line is the linear floor. (a) Inference cost per sequence: E24x's exclusive-node sweep in L ($B = 512$, $N_g = 401$, 8 BLAS threads); every BP-based arm shares grid BP's cost, and the network costs are those of E24's largest screened architectures, upper bounds for the smaller ones E33 selected. (b) Training wall clock per fit as E33 recorded it (shared nodes, indicative). Regenerated by `tools/make_field_runtime_fig.py`.

Figure 7 is the comparison the note's section 9.8 asks to be measured rather than asserted. Grid BP costs 0.5, 1.0 and 2.0 ms per sequence at $L = 32$, 64, 128, linear in L, and the networks 4 to 320 μs: a network is 6 to 120 times cheaper per posterior mean. In accuracy the order is reversed and the gap is larger: the site energy sits at 3–4 % of the floor on every lattice, the spline at 6–7 %, while the best network is at 48 % and the window head climbs to the floor as h falls, as the note expected for a fixed window. The runtime–accuracy frontier therefore has two ends and nothing between them: the networks own the cheap end, the structured kernels the accurate one, and the mixture sits on the floor at BP's cost. The note guessed that EM-BP might be cheaper to train though slower per sample; on training the guess does not hold here: at $M = 2048$ a network trained in 9 to 22 minutes on these nodes and an EM fit took 35 minutes to five hours. Memory, the last item on the note's list,

was measured by E38 (peak resident memory of one call, sampled during the call, each in a fresh process): grid BP needs about five message-sized arrays, 266 MB per posterior-mean call at $B = 512$, $L = 32$, $N_g = 401$, linear in B, L and N_g (slopes 0.96, 0.86, 0.99; both predictions confirmed), against 2 to 147 MB for the six networks; an EM E-step stays at about 250 MB whatever M, because it runs in chunks of 256 chains, while a network training step takes 2 to 66 MB. A Linux node, sampling `/proc/self/statm` instead of macOS's `task_info`, gives the same verdicts (slopes 1.01 and 0.99; grid BP 257 MB against at most 61 MB) and measures the two cells the laptop could not hold: 2.0 GB at $L = 64$ and 4.0 GB at $L = 128$ for $B = 2048$, still linear.

6.5 The clean-data interface (E34, P2 and P3)

Table 9: E34's clean-data fits at $M = 2048$, sixteen seeds. "parked": fits whose first-site law has more than half its mass beyond $|a| = 3.5$ (amendment 2); "eig.": fits whose invariant law the eigensolve could not return (the estimator then used $\mathcal{N}(0, 1)$).

	mixture / spline		spline	Legendre tilt		
cell	ratio	> 1	risk	risk	parked	eig.
$\Theta = 0.3$, $h = 1$	53.4	16/16	0.0054	0.1026	5/16	0
$\Theta = 0.5$	20.2	16/16	0.0062	0.2455	5/16	2
$\Theta = 1$	8.4	16/16	0.0072	0.0146	1/16	0
$\Theta = 2$	6.5	16/16	0.0065	0.0051	0/16	0
$\Theta = 0.3$, $h = 0.5$	46.7	16/16	0.0039	0.1451	13/16	3
$\Theta = 0.3$, $h = 0.25$	38.7	16/16	0.0027	0.1458	14/16	2

P3 holds on every cell and every seed: with the channel removed, the mixture still sits 6.5 to 53 times above the spline, so what limits it is the linear drift, not the noise. P2 fails as written. Clean data help the spline and the MDN (clean over noisy risk about 0.4–0.5, nearly 16/16), but not the mixture (0.90–0.99: both fits share one biased limit) and not the Legendre tilt, whose clean fits are four to nine times *worse*: 38 of its 96 clean fits at $M = 2048$ park their law outside the data (186 of 384 over all sizes), the failure of N-33 at scale. Nor does the gap shrink with M for the spline: the one-view channel costs it a factor of two to two and a half in risk at every size. Reading the failures turned up a second face of the parking: some fitted Legendre kernels are *reducible* on the grid, a bulk and a pocket with no transition between them, so that two different laws are exactly invariant; the eigensolve then cannot return a unique one, and on 20 of the 384 clean fits it raised. The spline (768 fits, clean and noisy) and the site energy (384) show neither in any confirmatory fit.

6.6 Why E33's kernel shards failed, and amendment 3

All four kernel jobs died on 2–3 October on the same cell, $\Theta = 0.3$, $h = 0.25$, $L = 128$ at $M = 2048$. Reproduced on the deployed code, the culprit is one Legendre fit, parked in the extreme: from $a = 3$ the kernel steps to the grid edge with spread 0.002, $\log K$ spans 1133 nats, 3.9 % of its entries underflow to zero in the E-step, and the forward and backward beliefs stop overlapping. The mixture and the spline complete the same cell. Before any kernel row was read, amendment 3 fixed the rule: a fit that raises in EM is a failed row, counted against every prediction it touches and never for one; the harness now records it instead of stopping. The rerun reused the 76 cells that had finished (the new code gives bit-for-bit the same rows on a test cell) and completed on 4–5 October. It recorded 39 failed fits, every one the Legendre tilt on that cell (3, 6, 14 and 16 of 16 at $M = 32$, 128, 512, 2048); no other arm failed anywhere.

6.7 E35: a global latent and its repair

The consolidated note's section 8 predicts that a latent shared by a whole sequence leaves a first-order kernel with a bias that data do not remove, and that enlarging the state removes it. E35 tests both on a case with a closed-form exact score: $X_i = Gb + U_i$, with $G = \pm 1$ a fair sign per sequence and U the Gaussian linear chain at $\rho = 0.85$. The repair is the same mixture kernel on the state (X_i, G), the two phases tied by reflection, with the same number of parameters (`markov_score/augmented.py`).

P2 and P3 hold: at $b = 2$ the first-order risk moves by +6 % and +3 % when M quadruples while the augmented arm's falls by 58 % and 73 %, and the ratio climbs with b to 8.6 and 17.9, the augmented arm sitting 5.4 and 11.5 times below the linear floor. P1 fails as written: at $b = 0$ the two arms agree on average, but at this small risk either one moves by a factor of three between seeds, and a 10 % band cannot hold at two seeds. Not predicted, and the sharpest number: under the latent the paper's first-order estimator is 1.5–1.7 times *worse* than the linear floor at both sizes.

Table 10: E35, score risk against the closed-form exact score (seed 0 / seed 1). The linear floor is the LMMSE of the true covariance $b^2\mathbf{1}\mathbf{1}^\top + \Sigma_U$; at $b = 0$ it is the exact score.

b	M	first-order	augmented	linear floor	first / augmented
0	512	0.0110 / 0.0276	0.0028 / 0.0342	0 (exact)	3.87 / 0.81
0	2048	0.0102 / 0.0074	0.0111 / 0.0046	0 (exact)	0.92 / 1.62
1	512	0.0594 / 0.0611	0.0183 / 0.0140	0.0355 / 0.0352	3.25 / 4.36
1	2048	0.0594 / 0.0592	0.0072 / 0.0067	0.0355 / 0.0352	8.19 / 8.79
2	512	0.0930 / 0.0923	0.0269 / 0.0196	0.0617 / 0.0612	3.46 / 4.70
2	2048	0.0988 / 0.0955	0.0114 / 0.0053	0.0617 / 0.0612	8.64 / 17.92

the note asks for	where it is
four arms across Θ, h, σ	E33: learned pair energy (two capacities), closure = covariance twin (§3), window head, oracle score (grid BP, true kernel); twelve levels t; six cells
exact transfer-matrix sampling	`field.py`: eigensolve on 2401 points, continuum rejection sampler and lattice inverse-CDF sampler, agreeing
score error, posterior-mean error	E33's estimand; the posterior-mean error is the same quantity per level up to e^{-t}/Δ_t
$C(r)$, ξ, kinks, domains, bimodality	E34 `physics_truth` and `physics_learned`
generated-sample quality	E34 `physics_learned`; the direct sampler of E27
runtime, memory (§9.8)	E24x (uncontended reruns) and E38: Figure 7; the field adds nothing to BP's cost, which depends on N_g, L, B only
clean-data learning (§16.1)	E34 `clean`
symmetric W_θ, reversibility	by construction: every tilted walk is reversible (Proposition 7); the symmetric site energy is learned directly by `em_pair_quartic` (§2.1)
Θ read from the data	Corollary 8: $\hat{\Theta} = \hat{\gamma}s^2/h$ from any kernel of the class; P8
FFT transition (§9.2, §29 step 5)	E40: implemented (`bp_fft`); second-order accurate (6.7×10^{-4} from dense BP at $N_g = 401$) and 5.8× slower on one core there; tuned, it breaks even near $N_g = 3200$
posterior sampling (§9.7)	`markov_score.ffbs`: forward filtering, backward sampling on the grid, tested against enumeration
"minimal decisive experiment"	E32 (floor across Θ, h) plus E33's window arm
figures-first (§28)	all seven and a summary figure: `reports/figures-first-2026-10-06.pdf`
Gaussian score / Kalman smoother (§29 step 3)	the paper's appendix B, a proposition with its proof; measured by E39 (grid BP within 1.9×10^{-13} of both)
notation; metric definitions (§29 steps 1, 2)	`docs/NOTATION.md` (code, paper, this report; thesis concordance); `docs/METRICS.md`
global latent and its repair (§8, fig. 7)	E35 (§6): bias measured, repair by state augmentation confirmed (P2, P3)
subsampling a fine lattice	E36 (§4.2): kernel K^s exact; P1, P2 confirmed

7 Alignment with the consolidated note

8 Questions for Jérôme and Marc

1. The morning's question (directed kernel or symmetric pair energy?) is answered by Proposition 7: they are one class. The remaining choice is the *form* of the pair energy. The Gaussian-bond-plus-site-energy form (3) is the whole class of discretised gradient SPDEs, and the quartic site energy is exactly specified for ϕ^4 with five parameters; a general symmetric $W(a, a')$ would also cover non-gradient couplings at the price of a two-dimensional function to learn. Is that generality wanted for the paper, or is the site-energy form the right level, with the tilt bases as the agnostic alternative?
2. Gaussian closure of the messages is not well-posed on the field (§3); the closure arm is the covariance twin. Does that match what Marc had in mind by "Gaussian closure" for this benchmark, or did he mean the Kalman smoother of the linearised dynamics?
3. The temperature sweep is at matched $\lambda_1/\lambda_0 = 0.85$ rather than at fixed h, so that the field is compared with the linear-chain laboratory at matched second-order structure. Agreed, or would a fixed-h sweep be more natural for the physics reading?
4. Theorem 2's ratio is still 0.75–0.95 at $t = 3$: the ϕ^4 constants are large. Is it worth a sentence in the paper, or only the figure?

A Numerics

Eigensolve. $A = W^{1/2}\mathsf{T}W^{1/2}$ on 2401 points of $[-12, 12]$ (physical units), symmetric, `scipy.linalg.eigh`; $\psi_n = W^{-1/2}v_n$. Components below 10^{-13} of the maximum are eigensolver noise and are carried as $\log\psi_0 = -\infty$ (stationary mass below 10^{-26}); a kernel column leaving such a state is the normalised T column. Spectral accuracy: $|\Delta\lambda_0|/\lambda_0 < 10^{-10}$, $|\Delta\rho_1| < 10^{-10}$ under halving the spacing and under widening the domain to

$[-16, 16]$. Completeness: $\sum_n c_n^2 = \mathrm{Var}_\pi(\phi)$ to 10^{-10}.

Samplers. Continuum: propose $\phi' \sim \mathcal{N}(\phi, \Theta h)$, accept with $g(\phi')/\max g$, $\log \psi_0$ by cubic spline on the resolved range; burn-in $\lceil \log(10^{-12})/\log \rho_1 \rceil$ steps from $\mathcal{N}(0, s^2)$. Lattice: inverse-CDF on the columns of the discretised kernel. Agreement in variance, lag-one correlation and kurtosis within Monte Carlo error at 3000 chains.

Fourth cumulant. $\mathbb{E}[a_i a_j a_k a_l]$ for $i \le j \le k \le l$ equals $\langle \psi_0 | \, a \, R^{j-i} a \, R^{k-j} a \, R^{l-k} a \, | \psi_0 \rangle$ with $R = \mathsf{T}/\lambda_0$, a triple sum over the leading 256 modes with $\Phi_{nm} = \langle \psi_n | a | \psi_m \rangle$; κ_4 by Wick subtraction. The diagonal reproduces the marginal excess kurtosis to 10^{-6}.

B Tests

33 new tests (59 pass in the files they live in, with the pre-existing ones): `test_field.py` (14: stationarity, reversibility, column mass, the tilted-walk identity, refinement, widening, completeness, two samplers, cumulant diagonal, BP as a denoiser, the iso-correlation solver), `test_tilted_kernel.py` (7: the Gaussian embedding, Fisher's identity by finite differences, M-step ascent and contract, clean MLE on the Gaussian chain and on the field, EM through noise), `test_bp_projected.py` (6: identity and LMMSE on wide grids for the Gaussian and Laplace chains, the truncation failure pinned, the field's near-flat backward message pinned), `test_stationary.py` (+2: eigensolve against power iteration; the wall), the kernel M-step contract (+4). The fast suite is unchanged (178 passed).